PAPER_742: Sombrero Galaxy MUGE — Dust Lane Drag Term D_dust

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Abstract

The Sombrero Galaxy (M104) presents a unique astrophysical challenge: its prominent equatorial dust lane exerts measurable drag on bulge dynamics and star formation near the central black hole ($\sim 10^9\ M_\odot$). This paper derives the MUGE for the Sombrero Galaxy incorporating the new D_dust environmental term, extending the standard UQFF formalism to include optically thick dust lane physics via the F_env operator.

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1. Introduction

M104 (NGC 4594), the Sombrero Galaxy, is a lenticular/Sa galaxy at 28 Mpc with one of the most prominent dust lanes in the local universe. Its structure features a massive bulge, a flat disk embedded in dust, and a $10^9\ M_\odot$ SMBH. The dust lane creates a unique gravitational environment where extinction, drag forces, and angular momentum exchange compete with bulge and halo dynamics.

Hubble parameters for M104: - M_visible $\approx 8\times10^{11}\ M_\odot$ - M_BH $\approx 10^9\ M_\odot$ - Distance $\approx$ 28 Mpc - r_galaxy $\approx$ 50 kpc - ρ_dust $\approx 10^{-20}$ kg/m³ (dust lane) - B $\approx 5\times10^{-6}$ T (typical galactic field)

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2. Sombrero Galaxy MUGE

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g_Sombrero(r,t) = (G·M)/r² · (1+H(z)·t) · (1−B/B_crit)
                + (G·M_BH)/r_BH²
                + (U_g1 + U_g2 + U_g3 + U_g4)
                + U_i
                + (Λ·c²/3)
                + (ħ/√(Δx·Δp)) · ∫(ψ·H·ψ dV) · (2π/t_Hubble)
                + ρ_fluid·V·g
                + (M_vis + M_DM) · (δρ/ρ + 3·G·M/r³)
                + D_dust
```

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3. D_dust — Dust Lane Drag Term

The novel term D_dust models the retarding force exerted by the optically thick equatorial dust lane on bulge stellar orbits and gas dynamics:

```
D_dust = −k_dust · ρ_dust · v_orbit² · A_cross / r

  k_dust   = dust-gas coupling coefficient (~0.5)
```

```
  ρ_dust   = dust lane mass density (~10⁻²⁰ kg/m³ for M104)
  v_orbit  = local orbital velocity (m/s)
  A_cross  = cross-sectional area of interaction (m²)
  r        = radial distance from center (m)
```

For the Sombrero bulge region (r ∼ 1–5 kpc):

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D_dust ≈ −5×10⁻¹² m/s²   (retarding contribution)
```

This is comparable in magnitude to the dark matter perturbation term, making it non-negligible for bulge kinematics near r < 3 kpc.

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4. Black Hole Contribution

The Sombrero's SMBH (M_BH ∼ 10^9 $M_{\odot}$) dominates within r < 1 kpc:

```
g_BH = G · M_BH / r_BH²
g_BH(1 kpc) ≈ 2.4×10⁻⁸ m/s²
```

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5. U_g Terms (Sombrero Configuration)

```
U_g1 = μ_dipole · B           (AGN magnetic dipole)
       μ_dipole = I·A·ω_spin ≈ 10⁻⁵¹ A·m²

U_g2 = B_super² / (2·μ_0)    (aether superconductor field)
       B_super = μ_0 · H_aether, H_aether ≈ 10⁻⁵ A/m

U_g3 = G·M_ext / r_ext²       (companion galaxy influence)

U_g4 = k_4 · ρ_vac,[SCm] · (M_BH/d_g) · e^(−αt) · cos(π·t_n) · (1+f_feedback)
       k_4 = 1.0
```

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6. Environmental Forcing for M104

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F_env(t) = F_dust + F_BH + F_cosmo

F_dust = D_dust / g_Newtonian     (dust drag fraction)
F_BH   = G·M_BH / (r²·g_Newtonian)  (BH contribution fraction)
F_cosmo ≈ H(z)·t                  (cosmological expansion)
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7. Full Equation Solutions

For standard bulge parameters (r = 5 kpc, t = 0): - g_Newton ≈ 4.5×10^{-10} m/s² - H(z)·t correction ≈ +0.35% - (1−B/B_crit) ≈ 1 (negligible for B « B_crit) - D_dust ≈ -5×10^{-12} m/s² (−1.1% correction) - g_Sombrero ≈ $4.5\times10^{-10} - 5\times10^{-12}$ + quantum + DM terms

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8. Astrophysical Significance

The D_dust term explains: 1. **Reduced star formation efficiency** in the dust lane compared to dust-free bulges 2. **Kinematic misalignment** of dust-embedded stars vs. outer halo stars 3. **SMBH growth rate suppression** through dust-mediated angular momentum transfer 4. **Mid-infrared brightness** excess near the dust lane (consistent with Spitzer observations)

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9. Conclusion

The Sombrero Galaxy MUGE extends UQFF to dust-dominated lenticular/Sa galactic environments via the D_dust term. The dust lane drag force amounts to ~1% correction on bulge dynamics but significantly affects the SMBH growth environment and inner kinematics. Future integration with Spitzer/JWST dust mass measurements can refine D_dust for M104 and analogous systems.
